

Johnny Dunn

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Summary

Full-stack engineer specializing in production AI/ML: RAG, LLM fine-tuning, classical deep learning, quantization, edge inference, and evaluation harnesses. Builds low-latency conversational voice interfaces with streaming STT/TTS, VAD, diarization, and turn-taking.

Professional Experience

Machine Learning Engineer

Independent Consulting - Contract | Remote | May 2019 – March 2026

- Built real-time edge AI surveillance on embedded SBC (RK3588, 3 NPUs) in Go for Aye Eye Global (3+ months). Multi-cam RTSP, YOLOv8n detection, facial recognition, local quantized LLMs for AI assistant, React UI, shipping 5k+ produced units.
- Built unified ML pipeline in Python for Linkby (6-month project) automating feature engineering and model training from SQL for advertising data, enabling parallel training of RF, SVM, Naive Bayes, and LR models with A/B testing framework.
- Improved AI text detection accuracy from 55% to 71% for Grapple Research (5-month contract) by fine-tuning GPT-3.5 on expanded, deduplicated dataset. Reduced training costs by 64% with truncation & summarizing based on article structure.
- Developed document matching tool for ImmenseX (4-month engagement) reducing initial prompt writing by 60% for devs. Deployed RAG pipeline using Weaviate, Neo4j (hybrid vector and graph-based search) for conversational memories.
- Achieved 77% sentiment analysis accuracy (random forest) trained on 10K texts for HereAfterLegacy.ai; built out initial MVP to scrape scripts from TV shows to recreate digital persona of 3 fictional characters (14-month contract).
- Trained neural style transfer models for public exhibit applying film-inspired filters to selfies for Edelman (3-month project).
- Created user research, animated wireframes, and functional demo for MVP at Notebrush (15-month engagement).

Senior Blockchain Engineer

Tilting Point | New York, NY | Nov 2021 – Feb 2023

- Developed ERC-20/721 smart contracts with upgradability / proxy patterns, built and deployed backend REST APIs for NFT marketplace, and internal airdropping tool. Rescued \$1M MATIC through proxy contract methods regenerating Gnosis safes.
- Architected blockchain state gRPC microservices, supporting 4+ games with 1000+ events/sec (on/off-chain throughput).

Software Engineer 2

eBay Inc. | New York, NY | Jun 2020 – Apr 2021

- Built REST APIs in Scala / Spring-Boot showing new ads in real-time from clicking selected item options (in under 1s latency).
- Handled E2E ML delivery infrastructure (K8s + Jenkins) enabling teams to deploy models into sandboxed UIs; accelerated deep learning team iteration by 50% allowing ads team to validate changes without waiting for releases from other teams.

Cognitive Implementation Engineer

Amelia.ai | New York, NY | Jun 2017 – Dec 2017

- Implemented features like PII redaction for HIPAA, SOC 2 compliance for AI assistant serving Fortune 500 medical suppliers.

Projects

AgentOS – TS agent orchestration; [guardrails](#) presets ([PII](#) redaction), graph-based, [multimodal RAG](#). github.com/framersai/agentos
Quarry – Fully offline notetaking app that organizes itself with NLP and can answer your questions. github.com/framersai/quarry
tenets – MCP server for coding agents to aggregate context for prompts with neural ranking, fully offline. github.com/jddunn/tenets
Restless - Hierarchical attention network model for anti-malware; 93% accuracy, [ClaMP](#) dataset. github.com/jddunn/restless

Education

Bachelor of Fine Arts in Game Design (May 2017) – Parsons School of Design

Awards

5th Place, AT&T National VR and AR Challenge, with THiNKIMMERSIVE team. (<http://npe100.com/> / mobile app), 2017.

Skills

Programming Languages	Python, TypeScript, JavaScript, Go, Scala, C#, SQL, Solidity
Frontend	React, React Native, Next.js, Redux, Vue, Nuxt, Tailwind, D3.js, three.js
Backend	FastAPI, Django, Node.js (Express, NestJS), gRPC, GraphQL, RabbitMQ, Kafka, Stripe
AI / ML	PyTorch, TensorFlow, LangGraph, LlamaIndex, RAGAS, DeepEval, Hugging Face, Transformers, sklearn, spaCy, Pandas, NumPy, OpenCV, CUDA, XGBoost, Stable Diffusion, DSPy, promptfoo
Databases / DevOps	PostgreSQL, SQLAlchemy, Supabase, Redis, Neo4j, Pinecone, Weaviate, MongoDB, Elasticsearch, AWS (Lambda, EC2, EBS, Bedrock), Docker, Kubernetes, Jenkins, GitHub Actions
Tools / Platforms / Other	Unity3D, Adobe Creative Suite, Electron, Capacitor, Android Studio, Quasar, Selenium, Puppeteer